Question. An online retailer reports that the time customers spend on its website per visit has a mean of 12.4 minutes and a standard deviation of 5.0 minutes. The distribution of visit times is skewed to the right. A marketing analyst randomly selects 50 visits. Find the probability that the mean visit time in the sample is between 11.5 and 13.5 minutes. Show all steps using the SPDC structure.

Detailed prompts

S – State:

1. Define the population (true) parameter in context.
2. Define the sample parameter in context.
3. What is 50? What does it represent?
4. What are 12.4 minutes and 5.0 minutes? What do they represent? What is the shape of the population distribution?
5. Write the problem (the probability you are asked to find using $\bar{x}$).

P – Plan:

1. Check randomness: Was the sample random? Why does randomness matter?
2. Check independence (10% condition): How many visits must there be in the population? Is $n < 10\%$ of N?
3. Check normality: The population is skewed. Is the sample size large enough for the Central Limit Theorem ($n \geq 30$)? What is the shape of the sampling distribution of $\bar{x}$?
4. Compute the mean of the sampling distribution of $\bar{x}$.
5. Compute the standard deviation of the sampling distribution of $\bar{x}$, and write the Normal model.

D – Do:

1. Write the sampling distribution of the sample statistic.
2. Write the probability statement in terms of $\bar{x}$.

3. Write the probability statement in terms of Z formula.
4. Convert both boundaries to z-scores.
5. Use the standard Normal table and write the value of the probability.

C – Conclude:

1. Start your conclusion with "There is about a ___ probability that…"
2. What is the sample size, and what is in the sample?
3. What statistic does the sample produce?
4. What interval is in the question?
5. What does this mean describe in context?